An overview of Vedic Mathematics

- Vedic Mathematics is a system of mathematical knowledge that originated in ancient India, particularly in the Vedas, which are ancient sacred texts of Hinduism.

- The term "Vedic Mathematics" was coined by Bharati Krishna Tirthaji Maharaja, a Hindu scholar, and mathematician, in the early 20th century.

Key features and principles of Vedic Mathematics

1. **Sutras (Mathematical Formulas):** 16 Sutras

2. **Correspondence with Modern Mathematics:** its principles can be applied to a wide range of mathematical problems, including arithmetic, algebra, geometry, calculus, and more.

3. **Sankhya Shastra:** an ancient Indian system of enumeration and calculation.

4. **Flexibility and Mental Calculation:** perform complex calculations mentally and with greater speed.

Sutras (Mathematical Formulas): 16 Sutras

1. **Ekadhikena Purvena:** By one more than the previous one.
2. **Nikhilam Navatashcaramam Dashatah:** All from 9 and the last from 10.
3. **Urdhva-Tiryagbhyam:** Vertically and crosswise.
4. **Paraavartya Yojayet:** Transpose and apply.
5. **Shunyam Saamyasamuccaye:** When the sum is the same, that sum is zero.
6. **Anurupyena:** Proportionately.
7. **Sankalana-Vyavakalanabhyam:** By addition and by subtraction.
8. **Puranapuranabhyam:** By the completion or non-completion.
9. **Chalana-Kalanabyham:** Differences and Similarities.
10. **Yaavadunam:** Whatever the extent of its deficiency.
11. **Shesanyankena Charamena:** The remainders by the last digit.
12. **Sopaantyadvayamantyam:** The ultimate and twice the penultimate.
13. **Ekanyunena Purvena:** By one less than the previous one.
14. **Gunitasamuccayah:** The product of the sum is the sum of the product.
15. **Gunakasamuccayah:** The factors of the sum are the same as the sum of the factors.
16. **Dhvajanka:** Flag.

1. Ekadhikena Purvena:

By one more than the previous one

$35^2 = 1225$

12 / 25

Part I- one more than the previous one

3+1= 4 3 X 4 = 12

Part II – (SECOND Number)^2

$5^2 = 25$

Nikhilam Navatashcaramam Dashatah:

All from 9 and the last from 10

100000 – 43658 = 056342

Step1: Need to subtract 5 digits, so separate 5 digits as one part, remaining is part two : 1 / 00000

Step 2: Subtract 1 from first part

Step3: Subtract first four digits from 9, last digit from 10

3. Urdhva-Tiryagbhyam:

Vertically and crosswise

24 x 36 = 864

2 4

3 6

Step1: Last digits : (right) multiply vertically

4 x 6 = 24 . keep 4 carry over 2

Step2: Cross product (2x6)+(3x4) = 24

keep 4 & add the last carry over

Step3: First digits: (left) multiply vertically and add the last carry over (2x3)+2 =8

4. Paraavartya Yojayet:

Transpose and apply

2587 ÷ 112

112	2	5	8	7
-1-2		-2	-4	
			-3	-6
	2	3	1	1

Quotient and Remainder - 23 and 11

No. of digits in quotient = (diff .b/t no.of digits in dividend and divisor) + 1

289487 ÷ 13103

13103	2	8	9	4	8 7
-3-10-3		-6	-2	0	-6
			-6	-2	0-6
	2	2	1	2	2 1

Quotent and Remainder - 22 and 1221

5. Shunyam Saamyasamuccaye:

When the **Samuccaye** is the same, that **Samuccaye** is zero

Meaning1: Samuccaye means common factor in all terms

e.g., 1: $12x-6x+3x+4x=0$

x is common factor just equal to 0

$x = 0$ is the answer

e.g., 2: $7(x+1)=8(x+1)$

$x+1$ is common factor just equal to 0

$x+1=0 \Rightarrow x = -1$

5. Shunyam Saamyasamuccaye:

Meaning2: Samuccaye means product of independent terms

e.g., 1: (x-7)(x-9) = (x-21)(x-3)

7x9 =63 21 x 3 = 63 **solution x = 0**

If the product of independent terms are same, then the solution is simply x=0

e.g., 2: (x+8)(x+9) = (x+3)(x+24)

8x9 =72 3x24 = 72 **solution x=0**

5. Shunyam Saamyasamuccaye:

Meaning3: Samuccaye means sum of the denomenators, when the numerators are same.

e.g., : $\frac{1}{3x-1}+\frac{1}{4x-1} = 0$

The numerators are same, Add the denominators and put =0

$$(3x-1)+(4x-1) = 0$$

$$7x - 2 = 0$$

$$x = 2/7$$

5. Shunyam Saamyasamuccaye:

Meaning4: Samuccaye means combination → if the sum of the denomenators is equal to the numerators then equate that sum to zero

e.g., $\frac{2x+5}{2x+11} = \frac{2x+11}{2x+5}$

Sum of numerators = 2x+5+2x+11 = 4x+16

Sum of denomenators = 2x+11+2x+5 = 4x+16

Equate that sum to zero 4x+16=0 => x = - 4

6. Anurupyena: Proportionately

e. g., 46 X 44 =

Working base: 40

Multiplication base = 10 x 4 = 40

Division = 100 / 2 = 50

46	+6
44	+4
cross add	Product
50	24 (keep 4 and carry 2)

x4 (mul.base)

200 +carry 2= 2024

7. Sankalana-Vyavakalanabhyam :

By addition and by subtraction

e.g., 1: Single digit add 43+ 8

43+10-2 = 53-2=51

e.g., 2: Double digit add 33+19

33+20-1 = 53-1= 52

e.**g., 3:** Subtract 55-9 = 55-10 +1 = 45 +1= 46

e.g., 4: 3 digit add 105+129

100+129+5= 229+5 = 234

8. Puranapuranabhyam :

By the completion or non-completion

1. Solve quadratic ($ax^2 + bx + c = 0$), biquadratic ($ax^4 + bx^2 + c = 0$)

e.g., 1: Qudratic equation: $x^2 + 2x - 8 = 0$

$$x^2 + 2x.1 + 1^2 - 1 - 8 = 0$$

$$(x+1)^2 - 9 = 0$$

$$(x+1)^2 = 9$$

$$(x+1)^2 = 3^2$$

$$x+1 = -3 \Rightarrow x = -4$$

x $\quad x + 1 = 3 \Rightarrow x = 2$

e. g., 2: Cubic Equation $x^3 + 6x^2 + 11x + 6 = 0$

compare $x^3 + 3.x^2.2 + 11x + 6 = 0$ & $a^3 + 3.a.b.(a+b) + b^3 = 0$

$$x^3 + 3.x.2.(x+2) + 2^3 - 8 - 3.x.4 + 11x + 6 = 0$$

$$(x+2)^3 - 2 - 12x + 11x = 0$$

$$(x+2)^3 - x - 2 = 0$$

$(x+2)^3 = (x + 2) \Rightarrow a^3 = a \Rightarrow a(a^2-1) = 0$

$\Rightarrow a=0,\ a=1,\ a=-1$

$a=0 \Rightarrow x+2=0 \Rightarrow x=-2$

$a=1 \Rightarrow x+2=1 \Rightarrow x=-1$

$a=-1 \Rightarrow x+2=-1 \Rightarrow x=-3$ Solutions $x = -2, -1, -3$

9. Chalana-Kalanabyham :

Differences and Similarities

Solve $x^2 - 2x - 4 = 0$

$D = b^2 - 4ac$

$= (-2)^2 - 4.1.(-4) = 20$

Differentiate => $2x-2 = \pm\sqrt{20}$

$2x-2 = +\sqrt{20}$, $2x-2 = -\sqrt{20}$

$2(x-1) = +2\sqrt{5}$, $2(x-1) = -2\sqrt{5}$

$(x-1) = +\sqrt{5}$, $(x-1) = -\sqrt{5}$

$\mathbf{x = 1 + \sqrt{5}}$, $\mathbf{x = 1 - \sqrt{5}}$

10. Yaavadunam :

Square its deficiency,Whatever the extent of its deficiency

Find the squares between 1 to 100

e.g.,: $94^2 = (94 - 6)^2$

$= 88 \mid 6^2$

$= 8836$

Find the squares more than 100

$102^2 = (102+2)^2$ $\quad$ $102 - 100 = 02$

$= 102 \mid 02^2$

$= 102\ 04$

Squaring from 969 to 999

$969^2 = (969 - 31) \mid 31^2$

$1000 - 969 = 31$

$= 938 \quad 961$

11. **Shesanyankena Charamena:**

The remainders by the last digit

Converting recurring decimal to fractions

	Quotient	Remainder	x7	Last digit
1/7				
10/7	1	3	21	1
30/7	4	2	14	4
20/7	2	6	42	2
60/7	8	4	28	8
40/7	5	5	35	5
50/7	7	1	07	

1/7= 0.142857

12. **Sopaantyadvayamantyam:**

The ultimate and twice the penultimate

Ultimate + Twice the penultimate (U+2P)

624 X 12 = -----

Step1: make a sandwich number with zero

0 6 2 4 0

P U

U+2P => (6+(2X0)) (2+(2X6)) (4+(2X2)) (0+(2X4))

=> 6 14 8 8 => 7 4 8 8

13.Ekanyunena Purvena:

By one less than the previous one

9999 x 2378 = 23777622

2377 / 7622

Part I- One less than 2378 is 2377

Part II – (9-2)(9-3)(9-7)(9-7) = 7622

14. **Gunitasamuccayah :**

The product of the sum of the coefficient is equal the sum of the coefficient in the product

$x^2+5x+6=0$

$(x+3)(x+2)=0$

coefficient of x^2 is 1

coefficient of x is 5

Constant coefficient is 6

sum of the coefficient is 1+5+6= 12 --(I)

Higher degree coefficient is 1, substitute 1 in factors
(1+3)(1+2) = 4x3=12 ---(II)

15. **Gunakasamuccayah:**

The factors of the sum are the same as the sum of the factors

$x^2+5x+4= (x+4)(x+1)$

$2x+5 = (x+4)+(x+1)$

The factors of the sum are the same as the sum of the factors.

16. Dhvajanka: Flag

Division 74862 ÷ 73

73 | $74_1 8_4 6\ _5 2$

$3\ 0\ 6\ 15$

$1\ 18\ 40\ 37$

quotient 1025

QUOTIENT = 1025
Remainder =37

Step1: 7/7= quotient 1

Step2: 3x1 =3

Step3: 4-3=1

Step4: 1 / 7= quotient 0, remainder 1

Step5: 3 x 0 =0

Step6: 18-0 =18

Step 7: 18/7= quotient 2, remainder 4

Step 8: 3 x 2 =6

Step 9: 46-6 =40

Step10: 40/7= quotient 5, remainder 5

Step11: 3x5=15

THANK YOU